

Pharmacological Directionality for Activity-Dependent Neural Network Rewiring During ECT, Psychedelics, and rTMS

Executive summary

Electroconvulsive therapy (ECT), classic/novel psychedelics, and repetitive transcranial magnetic stimulation (rTMS) can be conceptualized as **high-gain perturbation-plasticity interventions**: they transiently disrupt entrenched network dynamics (pathological attractors) and open a window in which synapses, intrinsic excitability, and inter-regional coupling become unusually modifiable. Evidence spans (i) seizure-/depolarization-associated transcriptomic and neurotrophic cascades in ECT/ECS, (ii) rapid structural synaptic remodeling and large-scale network desegregation under psychedelics, and (iii) stimulation-parameter- and state-dependent plasticity under rTMS that can be captured by spike-timing-dependent plasticity (STDP)-like models. ¹

A critical implication is that the plasticity induced by these interventions is **value-neutral**: it increases the brain's capacity to change, but it does not, by itself, specify *which* connections should strengthen, weaken, be pruned, or be stabilized. Direct experimental evidence for this “directionality requirement” is especially strong for psychedelics: psilocybin-induced connectivity changes are *network-specific* and depend on drug-evoked spiking activity—silencing a presynaptic region during drug administration disrupts the rewiring.

² Complementarily, the antidepressant “iPlasticity” framework argues that pharmacologically induced juvenile-like plasticity in adulthood becomes therapeutically meaningful only when paired with **training/rehabilitation/psychotherapy** that provides structured activity patterns for activity-dependent network refinement. ³ For rTMS, convergent evidence indicates that effects on networks are **brain-state dependent**, helping explain inter-individual variability and highlighting that concurrent cognitive/oscillatory states act as directional constraints on stimulation-induced adaptation. ⁴

Pharmacological treatments can, in principle, provide “directionality” by **biasing the rules and substrates of activity-dependent rewiring** across multiple scales: (a) shifting synaptic plasticity thresholds (metaplasticity), (b) altering neuromodulatory tone that gates Hebbian/STDP learning, (c) rebalancing excitation/inhibition (E/I) and interneuron maturity, (d) modifying extracellular matrix/perineuronal nets (PNNs) that stabilize circuits, (e) modulating microglial-mediated pruning and complement signaling, and (f) shaping transcriptional/epigenetic programs (e.g., BDNF/TrkB, immediate early genes, chromatin modifications) that consolidate—or constrain—new wiring. ⁵

Clinically and translationally, the key opportunity is **precision scheduling**: selecting drug classes that (i) open plasticity windows (“gates”), (ii) shape learning rules during the window (“steering”), and/or (iii) stabilize adaptive changes after the window (“consolidation”), while managing safety (seizure threshold, manic switch risk, cardiovascular effects, drug-drug interactions) using existing neuromodulation and pharmacology guidelines. ⁶

Conceptual framework for directionality

“Directionality” here refers to **biased convergence** of activity-dependent rewiring toward functional, symptom-reducing network configurations rather than random drift or reinforcement of maladaptive motifs. Three ideas—each supported empirically—make the case that ECT/psychedelics/rTMS require directional guidance:

First, **activity-dependent plasticity is permissive, not prescriptive**. Across developmental and adult plasticity literatures, network rewiring typically depends on structured experience, training, or patterned activity; pharmacology can “re-open” a permissive state, but the environment determines what is learned. The iPlasticity account explicitly frames antidepressant-induced juvenile-like plasticity as enabling activity-dependent rewiring by training/rehabilitation cues. ³

Second, **neuromodulatory interventions amplify sensitivity to state and inputs**. rTMS effects are described as inherently state dependent (cognitive, oscillatory, and history-dependent), linking target engagement—and ultimately clinical efficacy—to the brain’s momentary activity regime. ⁴ Psychedelic clinical science similarly emphasizes extra-pharmacological determinants (“set and setting”) and the need to standardize/report them, consistent with the view that context and guided experience shape enduring outcomes. ⁷

Third, in psychedelics there is direct mechanistic proof that **plasticity can be sculpted by manipulating concurrent activity**. In a mouse brain-wide mapping study, psilocybin strengthened specific long-range input routing while weakening recurrent cortico-cortical loops, and silencing a presynaptic region during drug administration disrupted the rewiring—demonstrating that “what is active during the window” matters causally. ²

Within this framework, pharmacological treatments can supply directionality by shifting the *learning landscape*—plasticity thresholds, eligibility traces, synaptic tagging/capture, interneuron gating, and consolidation rules—so that the activity patterns occurring during and after ECT/psychedelic/rTMS windows preferentially produce adaptive network changes.

Diagram of the core logic and intervention points

flowchart TB

A[Pathological network attractor
(e.g., rumination loop, threat bias,
CSTC compulsion loop)] --> B[Neuromodulatory perturbation
ECT / Psychedelic / rTMS]

B --> C[Plasticity window opens
↑ excitability/entropy, ↑ BDNF/IEGs,
↑ synaptic remodeling potential]

C --> D{Directional constraints present?}

D -- No --> E[Underspecified rewiring
variable outcomes; maladaptive
strengthening possible]

D -- Yes --> F[Biased activity-dependent rewiring
Hebbian/STDP + neuromodulatory gating]

F --> G[Consolidation & stabilization
ECM/PNNs, sleep, homeostatic
plasticity]

G --> H[New attractor basin
symptom-reducing network dynamics]

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subgraph Pharmacological levers
  P1[Gate: reopen plasticity\n(SSRIs/iPlasticity, HDACi,\nECM/PNN modulators)]
  P2[Steer: bias learning rules\n(NMDA/AMPA modulation,\nDA/NE/ACh tone,\nTrkB facilitation)]
  P3[Stabilize: prevent relapse\n(anti-inflammatory/microglia,\nE/I and oscillatory stabilizers)]
end

P1 --> C
P2 --> F
P3 --> G

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The “gate-steer-stabilize” decomposition is consistent with mechanistic evidence that (i) antidepressants can induce permissive juvenile-like plasticity and require behavioral guidance (gate), (ii) neuromodulators gate timing- and salience-dependent synaptic plasticity rules (steer), and (iii) extracellular matrix/microglia contribute to stabilization and pruning (stabilize). ⁸

Global disruption and plasticity mechanisms in ECT, psychedelics, and rTMS

ECT as global disruption with multi-day plasticity cascades

Global disruption. Clinical ECT induces a controlled seizure; emerging work indicates that a second large-scale phenomenon may reliably follow ECT seizures—hemodynamic signatures consistent with cortical spreading depolarization (CSD) in humans—suggesting an additional “reset-like” wave whose dynamics depend on stimulation parameters and electrode configuration. ⁹ ECT also produces robust changes in large-scale connectivity (e.g., disruption and reconfiguration of resting-state networks), consistent with a system-level perturbation that can destabilize maladaptive coupling patterns, though mechanistic specificity remains under active investigation. ¹⁰

Enhanced neuroplasticity. In animal ECS models (often used as mechanistic proxies for ECT), seizures drive high-amplitude activity-dependent gene induction (immediate early genes, neurotrophic factors) and structural remodeling. For example, serial ECS alters dendritic complexity and increases spine density and cellular proliferation in dentate gyrus; *Egr3* influences dendritic remodeling components of this response. ¹¹ Transcriptomic shifts after ECS have been measured across thousands of genes, and epigenetic regulation of activity-dependent genes (e.g., *Arc*, *c-Fos*) after ECS has been demonstrated in rats. ¹²

Human evidence of plasticity-linked structural and synaptic changes. In humans, ECT has been associated with hippocampal/dentate gyrus changes in imaging studies, and hippocampal enlargement has been linked to cognitive side effects, underscoring that plasticity and structural change are not uniformly beneficial. ¹³ Post-mortem evidence in depressed patients suggests increased immunoreactivity of hippocampal neuroplasticity markers after ECT, supporting a plasticity-enhancing interpretation (with obvious limitations: post-mortem design, confounding by illness and treatment histories). ¹⁴ Synaptic density changes following ECT have also been probed with SV2A PET-MR (¹¹C-UCB-J) in late-life depression

in a longitudinal pilot study, indicating feasibility for direct synaptic-density tracking in humans (limitations: small samples, pilot design). ¹⁵

Psychedelics as transient network desegregation plus synaptogenesis

Global disruption/desegregation. Human neuroimaging studies show that classic psychedelics reduce between-network segregation and alter information flow—findings often interpreted as transient weakening of rigid network modularity. For example, a simultaneous EEG-fMRI study of DMT reported decreased between-network segregation, including effects involving default-mode and frontoparietal/salience systems. ¹⁶ Directed functional connectivity analyses have reported broad decreases in directed connectivity under LSD/psilocybin/ketamine-like conditions (method-dependent), consistent with altered hierarchical control signals. ¹⁷ Predictive-processing-inspired accounts (e.g., REBUS-like models) are frequently invoked to interpret these changes as relaxed high-level constraints; empirically, these models are better viewed as computational hypotheses than established mechanisms. ¹⁸

Enhanced neuroplasticity: structural and synaptic remodeling. Primary preclinical evidence shows rapid neuritogenesis/spinogenesis after serotonergic psychedelics. A foundational study reported robust increases in neuritogenesis and/or spinogenesis in vitro and in vivo with serotonergic psychedelics, highlighting “psychoplastogen” effects. ¹⁹ Psilocybin has been shown to increase synaptic strength (AMPA/NMDA ratios) in hippocampal pathways in mice, correlating with antidepressant-like behavioral measures. ²⁰ In a large-animal translational step, a single dose of psilocybin increased synaptic density (SV2A) and decreased 5-HT2A receptor density in pig brain autoradiography, suggesting persistent synaptogenesis with receptor downregulation (model limitations: species differences, dosing/route differences relative to human psychedelic therapy). ²¹

Direct evidence for activity-dependent, sculptable rewiring. A key mechanistic advance is the demonstration that psilocybin’s long-range connectivity changes are network-specific and depend on spiking activity during drug administration; silencing a presynaptic region disrupts the rewiring. ² This is unusually direct evidence that “directionality” can be imposed by manipulating endogenous activity patterns during the psychedelic window.

Classic vs “novel” psychedelics and dissociation of subjective effect from selected biomarkers. Pharmacological blockade studies suggest partial dissociations between subjective psychedelic experience and some biological readouts. In a placebo-controlled human study, the 5-HT2A antagonist ketanserin reduced LSD subjective effects and adverse cardiovascular effects, while not affecting LSD-associated elevations of BDNF in that experiment (interpretation caveat: peripheral BDNF is an indirect proxy for central plasticity and can be influenced by multiple confounds). ²² This motivates experimental strategies to test whether plasticity-enhancing cascades (or their proxies) can be preserved while modulating phenomenology—and whether that impacts durable therapeutic rewiring.

rTMS as targeted perturbation with STDP-like and BDNF/TrkB-linked plasticity

Global vs targeted disruption. Compared with ECT and psychedelics, rTMS is typically more spatially targeted, yet it can drive network-level effects via connectome-mediated propagation. Modern accounts emphasize that clinical impact depends on target engagement within networks, and that effects are **state dependent** (cognitive state, oscillatory state, recent history), helping explain response variability. ⁴

Enhanced plasticity: synaptic and cellular mechanisms. A 2025 open-access study in mouse organotypic brain tissue cultures showed that 10 Hz repetitive magnetic stimulation induced long-term synaptic potentiation via coordinated pre- and postsynaptic activation; blocking BDNF/TrkB prevented potentiation, and TrkB activation could switch electrically induced LTD into LTP—supporting an STDP-like mechanistic account with BDNF/TrkB as a necessary component. ²³ This provides a concrete mechanistic bridge between stimulation parameters, timing rules, and neurotrophin signaling.

Gene expression and trophic effects. In rodents, rTMS has been associated with immediate early gene induction after single sessions and increased BDNF expression after repeated sessions, suggesting parameter-dependent transcriptional engagement. ²⁴ Systems-level neurometabolic changes have been measured in humans (and meta-analytically summarized) using MR spectroscopy, including post-rTMS changes in glutamatergic markers (e.g., Glx) in depression cohorts (limitations: heterogeneity of protocols, small samples, variable acquisition standards). ²⁵

Comparative snapshot of intervention-induced plasticity windows

Intervention	Principal “global disruption” phenomenon	Plasticity substrates engaged	Directionality evidence	Key limitations for mechanistic inference
ECT	Seizure + evidence for postictal CSD-like waves in humans; broad network perturbation ²⁶	IEG/BDNF induction, dendritic/spine remodeling, neurogenesis-like signatures (ECS) ²⁷ ; synaptic density measurable via SV2A PET pilot ¹⁵	Mostly indirect (clinical variability; cognitive side-effects linked to hippocampal enlargement) ¹³	Anesthesia, seizure threshold, electrode placement, and clinical heterogeneity; ECS≠ECT translation; cognitive trade-offs ²⁸
Psychedelics	Network desegregation, altered directed connectivity, increased signal diversity/entropy-like measures ²⁹	Rapid neuritogenesis/spinogenesis; synaptic strengthening (AMPA/NMDA) ³⁰ ; SV2A increases in pig ³¹	Strong: rewiring depends on activity during drug window (causal silencing disrupts rewiring) ²	Translation from rodent/pig to human therapeutics; separating pharmacology vs psychotherapy; adverse psychological reactions; drug–drug interactions ³²

Intervention	Principal “global disruption” phenomenon	Plasticity substrates engaged	Directionality evidence	Key limitations for mechanistic inference
rTMS	Targeted perturbation propagating through networks; strong brain-state dependence ⁴	STDP-like synaptic rule engagement; BDNF/TrkB required in vitro ²³ ; IEG/BDNF transcription in rodents ²⁴	Strong for state dependence; mechanistic directionality inferred from timing/state dependence and associative stimulation paradigms ³³	Parameter heterogeneity; translating cellular models to clinical protocols; medication confounds; network targeting uncertainty ³⁴

Pharmacological levers that can bias rewiring across biological scales

This section organizes drug classes by *mechanistic bias points*—how they could alter the direction of activity-dependent rewiring during ECT/psychedelics/rTMS windows. Each mechanism is linked to representative agents and evidence type (human/animal/in vitro) with key limitations.

Biasing synaptic strengthening/weakening and plasticity thresholds (metaplasticity)

A central way drugs can impose directionality is by shifting the **thresholds** for LTP/LTD and the stability of synaptic changes (metaplasticity). Mechanistically, interventions that increase AMPA throughput, facilitate NMDA-dependent coincidence detection, or directly facilitate TrkB signaling can bias which active synapses become “eligible” for strengthening during the high-plasticity window.

NMDA modulators (coincidence detection & learning-rule tuning). D-cycloserine (DCS), a partial agonist at the NMDA receptor glycine site, has clinical trial evidence as a cognitive enhancer for fear extinction when paired with exposure therapy, consistent with a directionality role: it does not supply content but amplifies extinction-learning mechanisms if extinction sessions are successful. ³⁵ This property is particularly relevant to PTSD/OCD models where targeted extinction or response prevention is the desired “direction” of rewiring.

NMDA antagonists with rapid plasticity cascades (ketamine-like). Ketamine’s rapid antidepressant effects are strongly linked (in mechanistic accounts) to downstream AMPA engagement, BDNF translation, and synaptogenesis; eEF2 kinase suppression and BDNF translation have been implicated in rapid antidepressant mechanisms in animal models. ³⁶ High-quality reviews emphasize both synaptic/circuit mechanisms and clinical constraints (dissociation, abuse potential). ³⁷ Importantly for “directionality,” ketamine-like agents may open a plasticity window but still require structured experience/therapy or targeted neuromodulation to shape long-term network outcomes (a hypothesis consistent with “gate-steer-stabilize” framing rather than conclusively proven in humans).

AMPA potentiators (amplifying active inputs). Positive allosteric modulation of AMPA receptors is a plausible steering mechanism because it selectively amplifies activity-bearing synapses. TAK-653 has been studied as an AMPAR PAM, including translational work using TMS-evoked measures in rats and humans as circuit biomarkers for AMPA modulation. ³⁸ Mechanistically, AMPA engagement is also implicated as downstream necessity for ketamine-like antidepressant effects. ³⁹ Limitations include seizure risk with excessive AMPA activation and the fact that most AMPA PAM work remains preclinical or early translational.

Direct facilitation of BDNF/TrkB signaling (synaptic eligibility + consolidation). Multiple lines support a TrkB-centered view of antidepressant and stimulation-linked plasticity: antidepressants can directly bind TrkB transmembrane regions facilitating synaptic localization/activation in vitro/in vivo, and TrkB signaling is integral to iPlasticity. ⁴⁰ In the rTMS synaptic plasticity study, BDNF/TrkB blockade prevented rMS-induced potentiation, and TrkB activation altered the direction of electrically induced plasticity (LTD→LTP), explicitly demonstrating a pharmacological direction-switch in a controlled model. ²³ Small-molecule TrkB agonists (e.g., 7,8-dihydroxyflavone) can rescue synaptic plasticity and learning in animal models, although clinical readiness is low and off-target concerns persist. ⁴¹

Biasing structural remodeling: spine formation/pruning and synaptic stabilization

Psychedelics and structural plasticity. Psychedelics can drive spine growth and neuritogenesis rapidly, creating substrates for long-range rewiring. ¹⁹ In psilocybin circuit-mapping experiments, the rewiring pattern involves selective strengthening/weakening of input routes, demonstrating that structural remodeling interacts with network topology and ongoing activity. ²

Extracellular matrix (ECM) and perineuronal nets (PNNs) as structural “brakes.” PNNs are strongly implicated as inhibitors of adult experience-dependent plasticity: degrading CSPGs with chondroitinase ABC reactivates ocular dominance plasticity in adult visual cortex, and PNN degradation in amygdala can render fear memories susceptible to erasure by extinction—strong evidence that ECM state can determine whether activity-dependent updating is possible. ⁴² Thus, drugs that reduce PNN integrity or alter PV-interneuron/PNN maturation can be conceptualized as **plasticity gates**.

SSRIs and interneuron/PNN “dematuration.” Chronic fluoxetine reduces parvalbumin expression and PNN markers in adult frontal cortex interneurons, suggesting a shift toward a more juvenile-like inhibitory microcircuit state consistent with iPlasticity. ⁴³ The original demonstration that chronic fluoxetine can reinstate adult cortical plasticity (ocular dominance) provides a canonical example that pharmacology can reopen plasticity windows, but activity/experience is needed to guide recovery. ⁴⁴

Microglia-mediated PNN remodeling. Microglia can interact with PNN dynamics; in a Cell Reports study, microglia enabled disassembly of mature PNNs during ketamine treatment, positioning microglia as potential mediators of plasticity gating and stabilization. ⁴⁵ This suggests anti-inflammatory or microglia-modulating drugs could bias the balance between plasticity and stabilization, but translation to psychiatry remains incomplete.

Biasing E/I balance and oscillatory regimes

E/I balance influences which ensembles dominate during neuromodulation windows and can therefore bias network “solutions.” Psychedelics have been explicitly discussed in relation to E/I modulation and sensitive-period-like sequences (heightened excitability followed by inhibitory refinement), though much of this is

integrative review-level inference rather than direct causal evidence in humans. ⁴⁶ For rTMS, MRS studies report changes in glutamatergic markers after treatment, implying neurometabolic shifts that could reflect E/I rebalance. ²⁵

Drug classes with plausible E/I-direction roles include:

- **GABAergic modulators / “synaptic stabilizers”** (benzodiazepines, anticonvulsants): may reduce excessive excitation and potentially prevent maladaptive strengthening, but can also blunt plasticity induction. A comprehensive review of psychotropic drug effects measured with TMS reports that glutamate antagonists robustly attenuate plasticity measures, and multiple drug classes alter excitability/plasticity in distinguishable ways—highlighting the need to treat concomitant meds as mechanistic variables in rTMS trials. ⁴⁷
- **Mood stabilizers** (lithium, lamotrigine, valproate): lithium engages GSK-3 β and neurotrophin-related cascades with translational evidence for plasticity-relevant effects; lamotrigine reduces glutamate release and influences excitability; valproate has epigenetic (HDAC-inhibitory) actions that can reopen learning windows (see below). ⁴⁸

Biasing neuromodulatory tone for salience- and timing-dependent learning (STDP gating)

Neuromodulators (dopamine, norepinephrine, acetylcholine) can gate whether correlated activity leads to LTP/LTD and can influence persistence of plasticity. A key conceptual synthesis is that “neuromodulation opens the STDP gate,” i.e., neuromodulators can determine whether timing-dependent pairing rules lead to stable synaptic changes. ⁴⁹

Noradrenergic agents (β -adrenergic signaling). β -adrenergic receptor activation can stabilize LTP and memory; mechanistic work links noradrenergic signaling to heterosynaptic LTP stabilization and synaptic tagging/capture. ⁵⁰ Clinically, β -blockade has been used as a directional tool for memory reconsolidation interference: in a randomized trial in PTSD, propranolol given before memory reactivation sessions reduced PTSD symptoms, consistent with reconsolidation-based updating as a directional mechanism (limitations: reconsolidation effects can be boundary-condition dependent; psychotherapy protocol fidelity matters). ⁵¹

Cholinergic agents (attention and input prioritization). Acetylcholine can facilitate or depress synaptic transmission in cortical slices, and is central to attention and encoding-retrieval modes; newer work shows acetylcholine can shape hippocampal pathway dynamics, though not always in the simplified “prioritize entorhinal over CA3” manner. ⁵² Rapid-acting antidepressant-like synaptogenic effects have also been shown with scopolamine through mTORC1 signaling and spine synapse increases in prefrontal cortex, illustrating that cholinergic manipulation can intersect with the same plasticity machinery invoked for ketamine-like agents. ⁵³

Dopaminergic agents (reinforcement learning and synaptic eligibility). Dopamine modulates STDP and can shape which activity patterns are reinforced; mechanistic reviews and empirical studies support dopamine-dependent modulation of timing-dependent plasticity and learning. ⁵⁴ This suggests dopaminergic agonists/antagonists could bias which ensembles “win” during neuromodulation windows, but psychiatric translation requires careful disorder-specific modeling (dopamine can be pro-psychotic in vulnerable individuals).

Biasing transcriptional and epigenetic consolidation programs

ECT/ECS gene programs. ECS drives broad transcriptional changes and engages epigenetic modifications on gene promoters; this provides a plausible substrate for long-term consolidation of network changes after an acute perturbation. ⁵⁵

Epigenetic plasticity gates (HDAC inhibitors). HDAC inhibition has been linked to enhanced extinction and memory updating in animal models, and valproate has human evidence consistent with reopening critical-period-like learning: adult men receiving valproate learned absolute pitch identification better than placebo, interpreted as facilitated critical-period learning. ⁵⁶ HDAC-related modulation of fear extinction has also been demonstrated in mouse models, supporting epigenetic control of extinction-memory formation. ⁵⁷ Limitations include non-specificity of systemic HDAC inhibition, potential for maladaptive plasticity, and high clinical risk burdens (e.g., teratogenicity for valproate in reproductive-age populations).

Immediate early genes (IEGs) as activity-dependent selectors. Psychedelics and ketamine have been compared in brain-wide mapping of c-Fos expression, providing region-level signatures of drug-evoked activation that plausibly shape which synapses become eligible for remodeling. ⁵⁸ In the psilocybin rewiring study, rewiring correlated with patterns of c-Fos expression, further linking transcriptional activity markers to circuit-level remodeling. ²

Timing, scheduling, biomarkers, and safety

Timing interactions: before, during, after the neuromodulatory window

A clinically useful scheduling view is to align pharmacology with three phases inferred from mechanistic work: **priming (gate), intervention-window steering (steer), and post-window consolidation (stabilize).**

flowchart LR

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A[Priming days-weeks\nraise plasticity capacity\nlower inhibitory brakes\n(SSRIs/iPlasticity;\nHDACi; ECM/PNN modulation)] --> B[Intervention hours\nECT seizure / psychedelic acute state / rTMS session\n(steer with neuromodulators,\nNMDA/AMPA/TrkB tuning)] --> C[Consolidation days-weeks\nstabilize adaptive wiring\nsleep, anti-inflammatory,\nprevent maladaptive escalation]
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This is consistent with (i) chronic antidepressant effects that reopen plasticity windows requiring activity-dependent guidance, (ii) acute psychedelic rewiring that depends on activity during the drug, and (iii) rTMS plasticity that depends on coincident pre/post activity and BDNF/TrkB signaling. ⁵⁹

Practical hypotheses for timing (not clinical recommendations):

- **Chronic priming (weeks):** SSRIs or other iPlasticity-promoting agents may demature inhibitory interneurons/PNNs and increase adaptability, potentially making subsequent rTMS or structured psychotherapy during/after ECT more “instructional.” ⁶⁰

- **Acute steering (hours):** Agents that gate NMDA-dependent learning (DCS) or AMPA throughput (AMPA PAMs) could be administered close to the behavioral instruction period (exposure/ERP, cognitive training) that is paired with rTMS or scheduled within the psychedelic integration window, to bias which active synapses strengthen. ⁶¹
- **Reconsolidation-based steering:** Propranolol given before memory reactivation sessions illustrates a concrete schedule-dependent directional intervention for PTSD-like networks. ⁵¹
- **Post-window stabilization (days):** Anti-inflammatory or microglia-modulating strategies might reduce maladaptive pruning or consolidate beneficial changes, but human causal evidence in the context of neuromodulation windows is limited; still, ECS can interact with neuroinflammation in models, and anti-inflammatory augmentation has depression evidence in other contexts. ⁶²

Dose considerations as mechanistic variables

Because your question is research-oriented, dose is treated here as a **mechanistic manipulator** rather than a prescribing recommendation.

- In psychedelic human pharmacology, receptor-antagonist pretreatment studies (e.g., ketanserin + LSD) show that dose and timing can selectively alter subjective effects while leaving some biomarkers unchanged in that experiment, highlighting the possibility of factorial designs separating phenomenology, physiology, and durable rewiring outcomes. ²²
- In rTMS, dose can be operationalized as pulses/session, intensity, and total sessions; safety guidelines emphasize parameter limits and screening to reduce seizure risk and adverse events. ⁶³
- In ECT, dose is typically individualized relative to seizure threshold and electrode placement; practice guidelines stress optimizing efficacy while minimizing cognitive adverse effects and monitoring concomitant medications that alter seizure threshold. ⁶⁴

Biomarkers to monitor “directional rewiring”

A directional-rewiring program needs biomarkers that capture both *plasticity capacity* and *the direction of network change*. Candidate modalities include:

Connectomics and functional dynamics - Resting-state fMRI network connectivity and segregation (especially DMN/salience/frontoparietal interactions) is directly relevant for psychedelics (desegregation effects reported) and for neuromodulation response tracking. ⁶⁵

- Directed connectivity metrics (e.g., Granger causality / effective connectivity) have shown reductions during psychedelic states, suggesting altered top-down control signals. ¹⁷

Perturbational biomarkers - TMS-EEG measures can index propagation and network engagement; rTMS literature emphasizes that target engagement depends on state. ⁶⁶

Synaptic density / synaptogenesis proxies - SV2A PET (11C-UCB-J) is a leading in vivo synaptic density marker; it has been used in an ECT pilot study, and psilocybin has been shown to increase SV2A density in pig brain. ⁶⁷

- Human psychedelic SV2A PET evidence is still emerging; a preprint suggests setting may influence SV2A outcomes, aligning with directionality-by-context hypotheses but requiring peer-reviewed replication. ⁶⁸

Neurochemical and inflammatory biomarkers - MR spectroscopy (glutamate/Glx, GABA) can track neurometabolic shifts after rTMS, with evidence for post-treatment changes. ²⁵

- Peripheral inflammation markers (CRP, IL-6, TNF- α) have been investigated in ECT; reviews suggest convergence between clinical features and inflammatory/neuroplastic mechanisms, though findings are heterogeneous and not yet individually predictive. ⁶⁹

Transcriptional activity proxies - Immediate early gene mapping (c-Fos) differentiates psilocybin and ketamine activation signatures in rodents and correlates with rewiring patterns in psilocybin circuit mapping, providing a mechanistic bridge to targeted “activity sculpting.” ⁷⁰

Translational/clinical implications and safety concerns

Concomitant medication as a mechanism, not nuisance. A TMS literature review emphasizes robust effects of psychotropic drugs on cortical excitability/plasticity measured with TMS, and notes that how medications impact clinical response to rTMS remains incompletely tested—implying that pharmac-rTMS should be studied as mechanism-based combination therapy rather than uncontrolled confounding. ⁴⁷

ECT medication management and cognitive risk. ECT practice guidance highlights the importance of managing medications that increase seizure threshold (e.g., anticonvulsants, benzodiazepines) and monitoring cognitive adverse effects. ⁷¹ Hippocampal enlargement correlating with cognitive side effects emphasizes that neuroplasticity can carry cognitive trade-offs, reinforcing the “value-neutral” principle. ¹³ Ketamine as an ECT anesthetic can increase seizure duration and reduce electrical dose, but meta-analytic conclusions do not support enhanced antidepressant efficacy overall; thus, “more seizure” is not equivalent to “better directed rewiring.” ⁷²

Psychedelic drug-drug interactions and risk management. A systematic review of classic psychedelic drug-drug interactions documents interactions with SSRIs, antipsychotics, and other agents, underscoring the need for careful protocol design when combining psychedelics with pharmacotherapies intended to steer plasticity. ⁷³ Lithium-psychedelic combinations have been associated with adverse outcomes in case-report literature, reinforcing the need for mechanistic and safety stratification. ⁷⁴ Ketanserin blockade studies illustrate that targeted receptor antagonism can strongly modulate acute subjective and neural effects of LSD. ²²

Regulatory and clinical readiness context. Device-based neuromodulation is regulated with safety standards; rTMS safety guidelines were developed by consensus efforts promoted by the International Federation of Clinical Neurophysiology ⁷⁵ and published as expert guidelines. ⁷⁶ In the U.S., the Food and Drug Administration ⁷⁷ permitted marketing of TMS for depression (2008) and later for OCD; these regulatory milestones matter for clinical readiness comparisons when proposing combination protocols. ⁷⁸ (These regulatory details should be treated as jurisdiction-specific; local approvals differ.)

Tables of drug classes, mechanisms, evidence strength, and readiness

Drug-class comparison focused on “directionality” mechanisms

Evidence strength is summarized qualitatively (High/Moderate/Preclinical/Early) and refers to *directionality-relevant evidence*, not overall clinical efficacy.

Drug class (examples)	Primary directional mechanism(s)	Evidence models & key findings	Evidence strength for directionality	Clinical readiness for combination protocols
Antidepressants: SSRIs (fluoxetine, etc.)	Plasticity gate via iPlasticity; interneuron/ PNN dematuration; TrkB facilitation	Chronic fluoxetine reinstates adult cortical plasticity (ocular dominance) ⁷⁹ ; reduces PV/ PNNs in adult frontal cortex ⁸⁰ ; iPlasticity review argues training/ psychotherapy provides direction ³	Moderate-High (preclinical causal + clinical synergy patterns) ³	High (widely used; but interactions with psychedelics and seizure threshold in ECT must be managed) ⁸¹
NMDA modulators: D-cycloserine	Steer extinction learning by enhancing NMDA-dependent plasticity during exposure/ERP	Improves extinction outcomes when paired with exposure; effect moderated by quality of exposure session ³⁵	High for psychotherapy-linked directionality	High (adjunct trials exist; mechanistic rationale clear) ⁸²
NMDA antagonists / rapid-acting antidepressants: ketamine/ esketamine	Gate + steer via AMPA→BDNF/ mTOR synaptogenesis; opens rapid plasticity window	eEF2K→BDNF translation mechanism (preclinical) ⁸³ ; reviews synthesize synaptic/circuit mechanisms ⁸⁴ ; ECT anesthesia augmentation not consistently beneficial clinically ⁸⁵	Moderate (strong plasticity evidence; directionality in humans less direct)	High for clinical use; combination with neuromodulation requires safety governance ⁸⁶

Drug class (examples)	Primary directional mechanism(s)	Evidence models & key findings	Evidence strength for directionality	Clinical readiness for combination protocols
AMPA potentiators (TAK-653; ampakines)	Steer by amplifying active excitatory inputs; may bias STDP-like integration	Translational TMS-evoked biomarker study in rats/ humans supports target engagement metrics ³⁸	Early-Moderate (mechanistic plausibility; limited psychiatric outcome trials)	Moderate (experimental/ early translational)
Neurotrophin enhancers / TrkB agonists (7,8-DHF)	Steer + stabilize via TrkB activation and synaptic consolidation	TrkB agonist mimics BDNF and rescues plasticity/ learning in animal models ⁴¹	Preclinical	Low-Moderate (not established psychiatric therapeutics)
Psychedelics (psilocybin, LSD, DMT; “psychoplastogens”)	Gate + steer: rapid spinogenesis; network desegregation; activity-dependent input routing	Psychedelics induce neuritogenesis/ spinogenesis ⁸⁷ ; psilocybin rewiring depends on activity during drug window ² ; DMT reduces network segregation in humans ¹⁶	High for activity dependence (preclinical); moderate for definitive human circuit directionality	Moderate (rapidly evolving; protocolized psychotherapy and interactions remain critical) ⁸⁸
Antipsychotics / 5-HT2A antagonists (ketanserin; atypical APs)	Brake/ terminate psychedelic effects; potential stabilize vs psychosis risk; may blunt plasticity or steer away from hallucination-linked activity	Ketanserin reverses LSD subjective effects and blocks neural effects in connectivity studies ²²	Moderate (clear acute effect modulation; long-term rewiring consequences unknown)	High for acute management; uncertain for “therapeutic steering”

Drug class (examples)	Primary directional mechanism(s)	Evidence models & key findings	Evidence strength for directionality	Clinical readiness for combination protocols
Epigenetic plasticity gates (valproate/ HDAC inhibition)	Gate via chromatin permissiveness; may enhance extinction/ updating	Valproate improves adult learning in a “critical-period-like” paradigm ⁵⁶ ; HDAC-related extinction modulation in mice ⁵⁷	Moderate (human signal exists; mechanistic specificity limited)	Moderate (high safety constraints)
Anti-inflammatories / microglia modulators (celecoxib, minocycline; experimental complement inhibitors)	Stabilize/ normalize maladaptive pruning and inflammatory constraints; may influence PNN remodeling	Microglia prune synapses via complement in activity-dependent manner ⁸⁹ ; ECS modulates neuroinflammation in models ⁹⁰ ; celecoxib augmentation evidence in depression meta-analyses ⁹¹	Early-Moderate (directionality inference; mixed/ heterogeneous clinical outcomes)	High for some drugs (NSAIDs) but uncertain mechanistic targeting
Cholinergic agents (scopolamine; AChE inhibitors; nicotinic/ muscarinic agonists)	Steer via attention/input selection and mTOR-linked synaptogenesis (scopolamine)	Scopolamine rapidly increases mTORC1 and spine synapses in PFC ⁵³ ; ACh modulates cortical synaptic transmission ⁹²	Moderate (mechanistic for scopolamine; broader cholinergic steering less tested)	Moderate (drug-dependent)
Synaptic stabilizers / glutamate modulators (lamotrigine; NAC)	Stabilize by reducing glutamate-driven excitability and oxidative stress; may prevent maladaptive strengthening	Lamotrigine reduces glutamate-related excitability ⁹³ ; NAC targets glutamatergic dysfunction in OCD context ⁹⁴	Early (directional synergy with neuromodulation untested)	High (available), but steering hypotheses need trials

Open questions and experimental designs to test directional pharmacology

Despite substantial mechanistic groundwork, decisive human evidence that specific pharmacological strategies *bias the direction* of ECT-/psychedelic-/rTMS-induced rewiring remains limited. Below are research-ready experimental designs emphasizing causal inference, biomarkers, and timelines.

Factorial “activity sculpting” studies in psychedelics

Rationale. Psilocybin rewiring depends on activity during administration. ² This enables causal tests of whether pharmacological modulation (e.g., AMPA potentiation, NMDA gating, β -adrenergic modulation) interacts with controlled activity patterns (cognitive tasks, affective reappraisal, exposure paradigms) to steer durable network changes.

Design (preclinical-to-translational). - **Preclinical arm (mouse):** psilocybin $\pm$ candidate steering drug (e.g., DCS; AMPA PAM; β -adrenergic agonist/antagonist) $\times$ controlled circuit activation/silencing (optogenetic/chemogenetic) during the acute window. - **Outcome measures:** monosynaptic tracing / connectomics (as in the mapping study), two-photon spine tracking, slice electrophysiology, IEG mapping (c-Fos), and behavioral readouts relevant to depression/PTSD/OCD analogs. ⁹⁵
- **Timeline:** baseline; acute (hours); 1 day; 1 week; 4–6 weeks (persistence).

Translational human arm (healthy volunteers or patient cohort): - **Double-blind factorial:** psilocybin + psychotherapy-like task vs psilocybin + neutral task, crossed with a steering drug (e.g., propranolol for reconsolidation-style tasks; DCS for extinction tasks) timed to peak learning window (pre-registered). - **Biomarkers:** fMRI connectivity/segregation; EEG complexity/signal diversity; SV2A PET subset if feasible (building on SV2A PET clinical pipeline and ECT pilot precedent). ⁹⁶
- **Primary hypothesis:** steering drug selectively amplifies task-consistent network changes (directional shift) rather than global plasticity markers alone.

Pharmaco-rTMS trials explicitly modeling state dependence

Rationale. rTMS is state dependent; BDNF/TrkB can switch plasticity direction in controlled models. ⁹⁷

Design (clinical). - **Population examples:** depression (DLPFC network), PTSD (fear extinction networks), OCD (CSTC loops). - **Intervention:** rTMS protocol fixed; randomize adjunct drug that targets (i) plasticity gating (SSRI continuation vs washout), (ii) steering (DCS before exposure/ERP + rTMS session), or (iii) stabilization (anti-inflammatory adjunct in inflammation-stratified subgroup). - **Key methodological feature: brain-state control** during stimulation (task engagement or oscillation-synchronized rTMS) to reduce variance and test interaction terms (drug $\times$ state). ⁹⁸
- **Outcome measures:** symptom scales; TMS-EEG propagation; MRS glutamate/GABA changes; resting-state connectivity; cognitive side-effect monitoring where relevant. ⁹⁹
- **Timeline:** baseline; mid-course; end-of-course; 1–3 month follow-up to test consolidation vs transient effects.

ECT “gate–steer–stabilize” designs with synaptic-density endpoints

Rationale. ECT induces broad plasticity signatures; synaptic density can be tracked with SV2A PET pilot methods; cognitive trade-offs exist (hippocampal enlargement linked to cognitive side effects). ¹⁰⁰

Design. - Randomized adjuncts: compare (A) standard ECT, (B) ECT + chronic SSRI priming (iPlasticity gate), (C) ECT + anti-inflammatory strategy in high-inflammatory subgroup, and (D) ECT + targeted cognitive training during the ECT course (directional activity). - **Primary outcomes:** symptom response; cognitive outcomes; SV2A PET (baseline and after course); structural MRI (hippocampal subfields); inflammatory markers (CRP/IL-6). ¹⁰¹

- **Safety constraints:** strict medication management per ECT guidelines; monitoring of seizure threshold and cognitive adverse events. ¹⁰²

- **Central hypothesis:** adjuncts shift the *pattern* of synaptic density change and network connectivity in ways that predict durable clinical benefit with reduced cognitive cost—testing directionality rather than “more plasticity.”

Key open mechanistic questions

Evidence gaps that most constrain translation:

1. **Can pharmacology selectively enhance *beneficial* activity-dependent rewiring without increasing maladaptive plasticity (e.g., anxiety sensitization, manic switching, cognitive adverse effects)?** The hippocampal enlargement–cognition link in ECT highlights this risk. ¹³
2. **Which “plasticity proxies” are actually causal mediators?** Peripheral BDNF can dissociate from subjective psychedelic effects in antagonist studies, raising questions about which biomarkers truly index direction-relevant plasticity. ¹⁰³
3. **How do ECM/PNN remodeling and microglial pruning interact in adult psychiatric circuits under these interventions?** PNN biology strongly gates adult plasticity, and microglia can mediate both pruning and PNN remodeling, but psychiatry-specific causal tests are scarce. ¹⁰⁴
4. **What is the optimal timescale for behavioral instruction relative to each intervention?** Psychedelic rewiring depends on activity during dosing; rTMS depends on brain state at stimulation; ECT’s effective “instruction window” is least characterized and may involve multi-day cascades. ¹⁰⁵

A unifying experimental imperative is to treat **concurrent activity patterns and pharmacology as first-class causal variables**, rather than assuming neuromodulation alone “repairs” networks.

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